To get access to an SD card, we need some hardware in the form of a breakout board or shield that will connect the card to the appropriate pins on the Arduino board. Adafruit Industries makes a nice compact microSD card breakout that can be wired up to the Arduino with only a handful of wires, while SparkFun Electronics has an inexpensive shield available that plugs directly into the Arduino board. Instead, we are going to use and recommend the Arduino Ethernet Shield, which in addition to having the hardware necessary for accessing a microSD card, also has Ethernet hardware built-in for future projects that open up the possibility for exploring Arduino Internet connectivity. The Arduino Ethernet Shield is convenient to use in that it plugs right into the pin headers on the Arduino board while still giving us access to all of the I/O pins not used by the devices on the shield.

In addition to the Ethernet Shield, we need to have a properly formatted microSD card as well. To read up not only on the procedure for formatting the SD card but also some of the differences in the hardware, check out the SD Card Notes page on the Arduino web site at `http://arduino.cc/en/Reference/SDCardNotes`. Once we have the hardware in place and a properly formatted microSD card, it's time to do something with it.

Because just storing data is not that interesting in and of itself, we will use the circuit shown in Figures 9-12 and 9-13 to create a very simple data logger to record an analog sensor input to a text file on the SD card called `data.txt`. We should start by mentioning a couple of things about the SD card on the Ethernet Shield. The Ethernet device and the SD card are both devices that use the Serial Peripheral Interface or SPI. We won't be getting into much detail with this protocol in this book, but fortunately for us, the SD library hides much of this interface from us. The Ethernet Shield connects digital pins 11, 12, and 13 to each SPI device with an additional pin needing to be tied to a hardware select pin to enable each device. The Ethernet device is connected to pin 10, the default for one SPI device, so the SD card has been connected to pin 4. In our code, we need to configure pin 10 as an output even though we are not using the Ethernet part of the shield for this example.

For the rest of our hardware, the sensor we will be using is the TEMT6000 breakout board from SparkFun to measure ambient light levels, although any other analog sensor would work as well. We will also use the Serial Monitor so that we can see the status of opening and writing to the file on the SD to tell us if something has gone wrong, what data is being written to the card, and when the file has been closed. Every time the Arduino is reset, it will open the file on the card and continue to update the values on the card for another round.

Once we know we can write files to the card, we will try out another sketch that reads values from the same text file on the card and outputs those values to a simple LED connected to a PWM pin—basically, a larger version of our Flicker Sketch from the last chapter. Because the SD card is connected to pins 4, 11, 12, and 13, and we can't use pin 10, we will connect our single LED to pin 9. Like the rest of the example code in this chapter, however, these sketches are not meant to be full-fledged projects and this is especially the case with the SD library. Think of these examples as proof-of-concept ripe for future exploration and expansion. So, let's get started.